

SWINBURNE UNIVERSITY OF TECHNOLOGY

DATA AND STRUCTURE

Unit Code: Computing Technology Project A (COS40005)
Project: Internal Assistant for Enterprise Knowledge (Slack chatbot with RAG)
Client: Katalon, Inc. (replacing the original VIB client)
Team: Group 7
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Table of Contents

A. Purpose of the Document....3
B. Question 1: Real Data for the Knowledge Base....3
2.1 Why the suggested “test data” lists do not fit....3
2.2 Recommended real sources....3
2.3 Why Katalon is easier than the original VIB brief....4
2.4 Using real edit history to test freshness....4
C. Question 2: Slack Structure and Channels....5
3.1 What Katalon has published....5
3.2 What is not public....5
3.3 Industry reference (not Katalon-specific)....5
3.4 Impact on the design....5
D. Question 3: Confluence and S3 Structure....5
4.1 Public evidence that Katalon uses Confluence....5
4.2 What is still unknown....6
4.3 S3 structure (provided by the client)....6
4.4 Implications for the team’s pipeline....7
E. Consolidated Questions for the Client....8
F. Summary and Next Steps....9
References....10

A. Purpose of the Document

This document summarises the research Group 7 has completed so far for the Katalon internal-knowledge Slack assistant. It answers three questions the team needs settled before building the test environment:

1. **Data:** which real (not mock or AI-generated) documents can fill the test Confluence site?
2. **Slack structure:** how does Katalon organise Slack, and how many channels are there?
3. **Confluence (and S3) structure:** how is Katalon's knowledge organised, and what is still unknown?

For each question the document separates **what is evidenced** (with a citation), **what is not public**, and **what must be asked of the client**. Section E combines all open questions into one message for the client mentor, Mr. Phạm Tiến Thuận Phát. It complements the environment documents (Group 7, 2026a, 2026b).

B. Question 1: Real Data for the Knowledge Base

2.1 Why the suggested "test data" lists do not fit

Lists of public datasets for Katalon Studio (for example, the CURA demo site, weather and astronomy APIs, Kaggle and government CSV files) serve a different purpose: they are inputs for **automated software testing with Katalon's product**. This project instead needs **written internal-knowledge pages** (policies, processes, product documentation) that employees ask questions about. Tabular or API data does not resemble Confluence content, so it cannot measure answer accuracy or freshness.

2.2 Recommended real sources

The sources below are real documents written by real companies and published openly. They are mapped to the Confluence spaces the client shared (Figure 1).

Target space	Real source	Why it fits	Licence / condition
Deal Desk	GitLab Handbook – Deal Desk	A real deal-desk team that streamlines opportunity management and is the first point of contact for sales support (GitLab, n.d.-a)	Creative Commons; free to copy and use with attribution (GitLab, 2016)
Partner Program	GitLab Channel Partner Program handbook	The primary handbook page for prospective and existing channel partners, with program guide and deal registration (GitLab, n.d.-b)	Creative Commons, attribution (GitLab, 2016)
Intranet / HR	GitLab Handbook (people policies) and 37signals Employee Handbook	Real policies and benefits; the GitLab handbook alone exceeds 2,000 printed pages (GitLab, n.d.-c)	37signals policies are licensed under Creative Commons Attribution (Basecamp, n.d.)
True Platform	Katalon Docs (including legacy release notes)	Katalon's own real, partly outdated product documentation (Katalon, n.d.-a)	Restricted: documentation use is limited to a reasonable number of copies (Katalon, n.d.-b); use a small excerpt and ask the client

Target space	Real source	Why it fits	Licence / condition
(Reference only)	Apache Software Foundation Confluence (cwiki.apache.org)	A live Confluence instance showing real space and page hierarchies	Apache License 2.0 (Apache Software Foundation, 2025)

Table 1. Real data sources mapped to the Katalon spaces

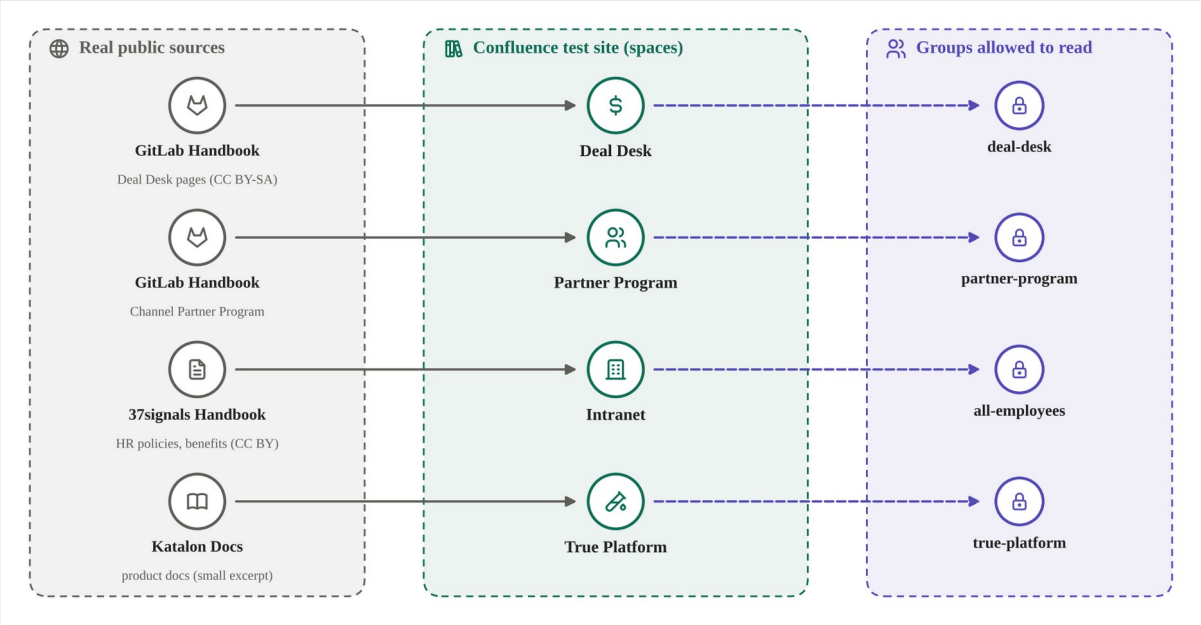


Figure 1. How real public sources fill the test Confluence spaces and which groups may read them

2.3 Why Katalon is easier than the original VIB brief

- **Industry openness:** software companies routinely publish documentation, and some (GitLab, 37signals) publish whole internal handbooks, so realistic material of the same kind exists.
- **Banking confidentiality:** a bank's internal procedures and customer data are not released to external teams, and no bank publishes an internal handbook that could be borrowed.
- **Known structure:** the client has already shared Katalon's real groups, page IDs and spaces, so the team knows exactly what to reproduce.

2.4 Using real edit history to test freshness

The GitLab Handbook is maintained as a Git repository, so it carries years of **real edit history**. The team can replay real commits (for example, a policy that changed) into the test Confluence site and measure how quickly the bot's answer changes. This gives the freshness criterion (30%) evidence based on real changes instead of invented ones. All reused content must be attributed in the final report.

C. Question 2: Slack Structure and Channels

3.1 What Katalon has published

Katalon source	What it shows
Katalon Docs – Slack integration with Katalon True	Katalon's product sends notifications about test results to a specified Slack channel (Katalon, n.d.-c)
Katalon Docs – Slack plugin for Katalon Studio	The plugin supports both public and private channels (Katalon, n.d.-d); this describes how customers use Slack, not Katalon's internal workspace
Katalon careers pages	Staff collaborate in Jira and Confluence (Katalon, n.d.-e); Slack is not mentioned
Katalon Community forum	Katalon stated there was no Slack group for Katalon users (Katalon Community, n.d.)
BuiltIn company profile	About 501 employees, headquartered in Atlanta (BuiltIn, n.d.)

3.2 What is not public

No Katalon document describes its internal Slack workspace: the number of channels, naming conventions, or whether one workspace or Enterprise Grid is used. This is internal information that companies rarely disclose, so it must come from the client.

3.3 Industry reference (not Katalon-specific)

If the report needs an illustrative structure, it should be labelled as industry practice. Slack's own internal communications team describes company-wide announcement channels, channels for each department (for example #sales-announcements) and many small teams, plus an open #slack-ama channel where anyone can ask leaders questions (Slack, n.d.).

3.4 Impact on the design

The number of channels **does not change the architecture**. The bot receives questions by direct message or @mention, and access rights come from Confluence, not from Slack channels. The only design rule that depends on channels is that answers given inside a channel must be ephemeral (visible only to the asker). Slack Connect users from outside the company must be refused, like guests.

D. Question 3: Confluence and S3 Structure

4.1 Public evidence that Katalon uses Confluence

Evidence	What it shows
Archived Katalon documentation URL `docs.katalon.com/display/KD/Test+Integrations ` (Wikipedia contributors, n.d.)	`/display/<space key>/` is Confluence's URL format: Katalon's public docs once ran on Confluence with space key KD
Short link `docs.katalon.com/x/KQHR` in the official `katalon-recorder-samples` repository (Katalon, n.d.-f)	`/x/…` is Confluence's tiny-link format, a second confirmation

Evidence	What it shows
Katalon job posts	Writers record documentation tasks in Jira and Confluence (Katalon, n.d.-e)
Material shared by the client	The current spaces (Intranet, Deal Desk, Partner Program, True Platform), groups and page IDs

4.2 What is still unknown

The number of spaces, how permissions are set (space level versus page restrictions), how groups are managed, page volume and update frequency are not public and must be confirmed by the client.

4.3 S3 structure (provided by the client)

The team has **no access to Katalon's Amazon S3**, which is the correct practice for an external team, and access is not needed: in both environment options the team uses its **own AWS account and bucket** (Group 7, 2026a, 2026b). On 11 November 2026 the client shared the reference structure used for ingested Confluence pages, and confirmed that **each folder holds the current version of its page** (T. T. P. Phạm, personal communication, November 11, 2026). Figure 2 reproduces it.

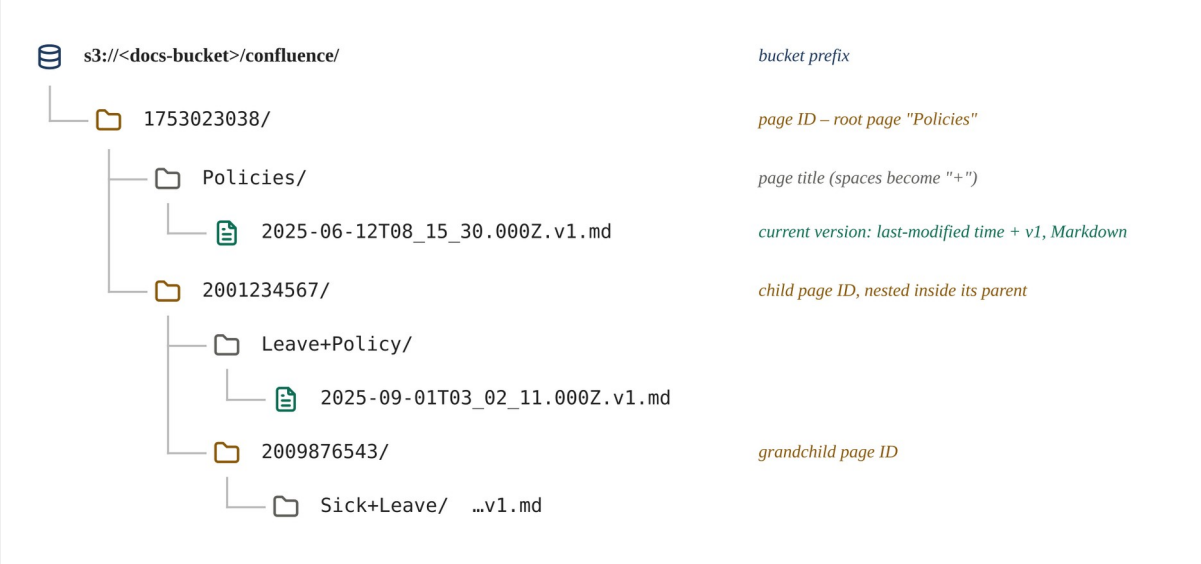


Figure 2. Reference S3 structure for ingested Confluence pages, as shared by the client

Element	Meaning	Use in the project
Page ID at the start of each folder	Identifies the Confluence page	Security: the permission filter works on page IDs
Child folders nested inside parent folders	The path encodes the page tree (ancestors)	Security: view restrictions on a parent page also apply to its child pages (Atlassian, n.d.), so ancestry matters
Title folder (spaces written as "+")	Human-readable page title	Accuracy: readable source names in answers
File name = last-modified time + .v1.md	Only the current version is kept, in Markdown	Freshness: compare with the time in Confluence; Markdown keeps headings and lists for chunking

Table 2. How to read the S3 structure

4.4 Implications for the team's pipeline

1. **Metadata sidecar files are needed.** Bedrock reads filterable attributes from a file named after the document with `.metadata.json` appended, stored next to it (Amazon Web Services [AWS], n.d.-a). The shared structure shows no such file, so the team's sync will write one per page, for example `2025-09-01T03_02_11.000Z.v1.md.metadata.json` with `page_id`, `ancestors`, `space_key`, `title`, `last_modified` and `allowed_groups`.
2. **Renamed or moved pages change the path.** Because the title and the parent chain are part of the key, a rename or move must delete the old folder before writing the new one; otherwise the bot may answer from duplicate or outdated content. This becomes a dedicated freshness test case.
3. **The space key is not in the path.** It must be read from the Confluence API and stored in the metadata.
4. **Deep page trees create long keys.** S3 object keys are limited to 1,024 bytes (AWS, n.d.-b), so the deepest page tree should be tested.

E. Consolidated Questions for the Client

The questions below can be sent to Mr. Phát as one message. Where a question is sensitive, an estimate or a yes/no answer is enough.

No	Question	Why the team needs it
S1	One Slack workspace or Enterprise Grid (several workspaces)?	Shape of the Slack sandbox
S2	About how many channels, and is there a naming convention (#team-, #proj-, #announce-)?	Realistic sandbox and channel tests
S3	Do Slack user groups match Confluence groups (e.g. deal-desk in both)?	Extra cross-check of who the asker is
S4	Is Slack Connect used with partners or customers?	External users must be refused like guests
C1	Confluence Cloud or Data Center, and which plan?	Correct API; whether Atlassian Guard exists
C2	Besides the four spaces shared, should any other space be included? Are personal spaces excluded?	Scope of the knowledge base
C3	Are permissions mainly at space level, or are there many page restrictions?	Complexity of the permission layer
C4	Are Confluence groups managed manually or synchronised from an identity provider through SCIM?	How users are matched across Slack and Confluence
C5	Roughly how many pages per space, and how often are they updated?	Sync schedule and embedding cost (freshness)
C6	Should attachments (PDF, Word, Excel) be answered from, or page text only?	Extraction work and cost
C7	Should archived pages and drafts be excluded?	Scope of the knowledge base
B1	Answered 11 Nov 2026 (Figure 2). Follow-up: what does “v1” mean – the Confluence version number or the export format version?	Correct freshness comparison
B2	Format is Markdown (answered). Is there a .metadata.json file per page, and which fields?	Permission filter depends on metadata
B3	What writes to the bucket, and how often?	Whether the team’s sync replaces or complements it
B4	Is the bucket already a data source for a Bedrock knowledge base?	Alignment with Katalon’s current design
B5	Could the team receive one or two redacted sample files or a sample metadata file?	Accurate test data format

Table 3. Open questions: S = Slack, C = Confluence, B = S3 bucket (B1 answered by the client)

F. Summary and Next Steps

Question	Answer so far	Next step
Data	Real, licensed sources are available for every space (Table 1)	Build a list of 30–50 pages with their space and group; confirm Katalon Docs use with the client
Slack structure	Not public; does not change the architecture	Send questions S1–S4
Confluence structure	Confluence use is evidenced; details are not public	Send questions C1–C7
S3 structure	Reference structure received from the client (Figure 2); metadata files still unknown	Send follow-up B1 and questions B2–B5

When the client answers, the team will update the Business Role Proposal (Section I) for its official version and the persona tables in both environment documents.

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